Positive Boundary Systems and Completely Positive Sewing: Physical Quotients, Finite Calibrations and Exact Obstructions

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Abstract

Boundary formulations of quantum gravity require more than nonnegative partition functions or positive scalar moments. A candidate theory must control positivity for noncommuting insertions, a reflection form on an explicitly chosen half-algebra, matrix-level positivity of cutting and sewing operations, normalization of operational instruments, descent through physical ideals, and compatibility under successive cuts. We place these requirements on a common typed boundary system, formulate relative completion ambiguity as noninjectivity of state restriction, and prove by elementary finite counterexamples that none of the relevant upgrades is automatic. In particular, positive scalar moments need not extend to a positive noncommutative functional; a faithful positive state need not be reflection positive; a positive, reflection-compatible map need not be completely positive; complete positivity need not imply reflection covariance, normalization, or quotient descent.

Two exact finite calibration families then show that the full collection of interfaces is mathematically consistent. The first is a doubled matrix-algebra construction with Kraus operators $\sqrt{\kappa}\,(\overline{F} \otimes F)$, a normalized failure outcome, canonical swap reflection, an explicit Osterwalder–Schrader form, inactive-block quotients, compatible finite dynamics, and associative composition. A two-packet instance gives closed posterior-distance and reconstruction-spectrum formulae. The second family uses finite Weyl algebras, Fourier charge sectors, group-averaged physicalization, neutral anchor dressing, and a matched-charge cut instrument. Choi, effect, quotient, and composition checks are analytic, with selected finite residuals reproduced computationally. These models are reference implementations of an operator-algebraic compatibility stack. Their labels and maps are selected rather than derived from a Virasoro-TQFT kernel, a CFT code, or a gravitational path integral. Accordingly, they establish finite consistency but supply no continuum, holographic, or quantum-gravitational promotion. We isolate the missing source comparison maps and the regulator estimates that would be required for such a promotion.

Scope and claim ceiling. This paper proves finite-dimensional statements and organizes standard operator-algebraic facts into a common boundary-sewing test. It does not construct physical gravitons, a continuum scalar field, a CEZ/Virasoro-TQFT sector, a CFT code, a large-N limit, or a theory of quantum gravity. "Physical quotient" below denotes a quotient whose algebraic descent is proved; a genuine gravitational interpretation additionally requires an independently supplied constraint or source bridge.

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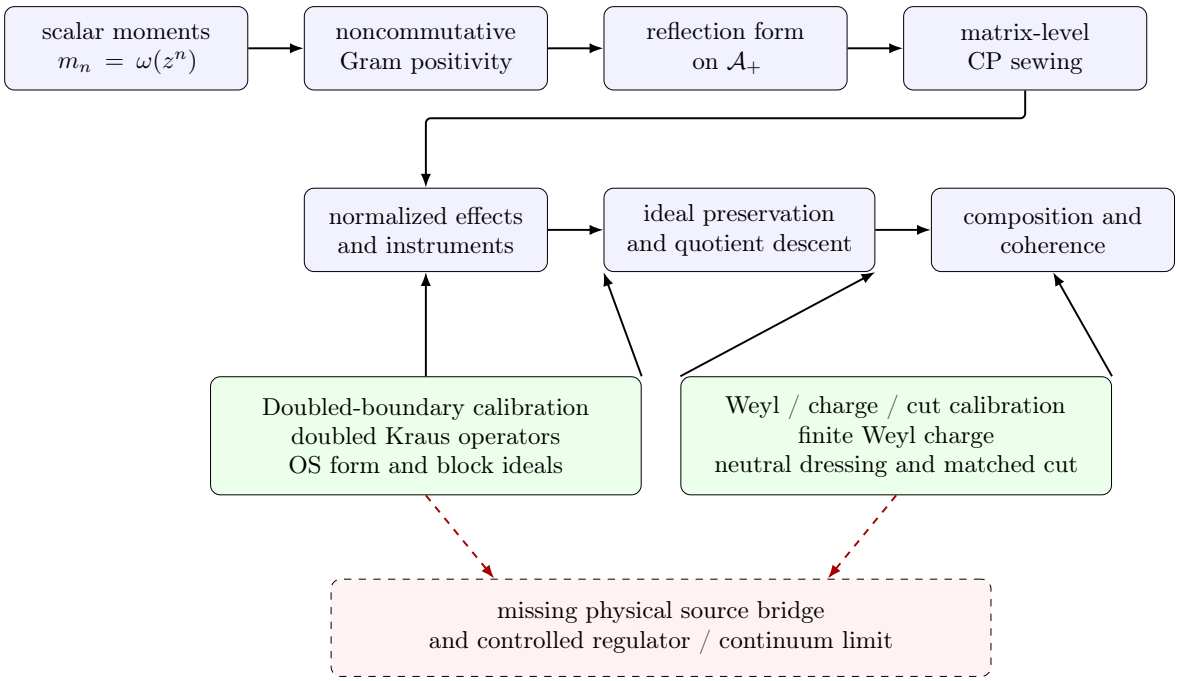


Figure 1: Graphical abstract. The horizontal and vertical boxes are distinct acceptance gates, not implications supplied by scalar positivity. The two exact finite models pass selected gates. Their dashed arrows end at an open source and continuum bridge.

1 Introduction

The gluing of quantum regions is often first described at the level of amplitudes. When the objects being glued carry gauge constraints, edge sectors, mixed states, or noncommuting observables, this scalar description is too weak. A nonnegative number for each closed region does not determine whether arbitrary operator insertions form a positive Gram matrix. A positive state on one boundary algebra does not determine whether a geometric reflection gives a positive reconstruction form. Even a positive map between two algebras can fail after tensoring with an untouched ancilla, which is precisely the stability demanded of a local physical operation. These distinctions are routine in operator theory and quantum information, but they are easy to blur when "positivity of sewing" is used as a single phrase.

The appropriate map variance is fixed throughout this paper. A cut $g : G \to C$ from a glued region to a cut presentation has a Heisenberg observable pullback

$$E_g : \mathcal{A}_C \longrightarrow \mathcal{A}_G, \qquad \Phi_g(\omega_G) = \omega_G \circ E_g. \tag{1}$$

Thus observables run from cut to glued, while states run from glued to cut. This separation

prevents a recurrent typing error: the state dual, a reverse inclusion, and a conditioned recovery rule are generally three different maps.

Complete positivity (CP) is the natural matrix-stable requirement on E_g. Stinespring's representation theorem and the Choi–Jamiołkowski criterion provide the basic structural and finite-dimensional tests [3, 11, 21]. Normalized collections of CP outcome maps are quantum instruments in the operational sense initiated by Davies and Lewis [6, 12]. Reflection positivity is a separate reconstruction condition, historically associated with the Osterwalder–Schrader axioms [15, 16]. General-boundary and positive formalisms already place amplitudes, probabilities, and composition on arbitrary boundaries [13, 14]. The aim here is not to rename these established ingredients. It is to state exactly how they must coexist on a common quotient and to give finite models in which their compatibility can be checked without interpretive shortcuts.

This need is especially sharp in gravity. Gauge-invariant gravitational observables are dressed and generally do not form naive commuting local subsystem algebras [8]. Holographic quantum error correction provides useful finite and code-subspace analogies [1, 10, 17], but an arbitrary qudit code is not thereby a bulk gravitational theory. In three dimensions, Virasoro TQFT and the Ponsot–Teschner fusion data provide concrete source structures [4, 5, 18]; a hand-chosen finite Kraus family is not a compression of those structures until a quantitative comparison is supplied.

The paper combines a typed boundary constitution with three bounded finite contributions. First, it defines a hierarchy of scalar, noncommutative, reflection, operational, quotient, and coherence tests with fixed map variance. Second, it gives exact 2×2 counterexamples showing that the main upgrades are logically independent. These examples are pedagogical and no priority claim is made for their elementary ingredients. Third, it presents two exact finite calibration families. One uses doubled Kraus operators and a canonical swap reflection; the other uses finite Weyl charge blocks, neutral anchor dressing, and a matched-sector quotient. The constructions are useful because invalid controls fail for explicit reasons and because every physical nonclaim can be localized to a missing map or limit.

The status vocabulary is deliberately conservative. A result marked STD is a standard theorem or consequence. A result marked DERIVED is proved here from printed finite hypotheses. A value marked COMPUTED is reproduced by a deterministic script at the stated parameters. An item marked CONDITIONAL depends on a hypothesis not established for the named source theory, and OPEN denotes a missing construction. This taxonomy matters because the algebraic consistency of a surrogate and its physical origin are different scientific questions.

2 The typed positive boundary system

Let Bdy be a category whose objects are admissible oriented boundaries and whose morphisms are admissible regions or cobordisms. Let Reg be a category or directed system of regulators.

Definition 2.1 (Positive boundary system)**.** A positive boundary system is a tuple

$$\mathfrak{Q} = (\mathsf{Bdy}, \mathsf{Reg}, \mathcal{A}, \omega, \Theta, \mathcal{S}, \mathcal{C}, \mathcal{D}, \mathcal{L}) \tag{2}$$

with the following typed components.

(i) $\mathcal{A}$ assigns a unital $*$-algebra $\mathcal{A}(\Sigma)$ to each boundary Σ, together with local subalgebras when a net is claimed.
(ii) ω is a compatible family of normalized positive linear functionals on declared test algebras.
(iii) Θ is orientation reversal or Euclidean reflection, with a specified positive-half algebra when OS reconstruction is claimed.

(iv) $\mathcal{S}$ is a family of gluing correspondences or maps with declared domain, codomain, and direction.
(v) $\mathcal{C}$ is constraint data: BRST/BFV differentials, Dirac constraints, averaging maps, or physical ideals.
(vi) $\mathcal{D}$ is dynamics or Ward data acting on the same physical structure.
(vii) $\mathcal{L}$ is a regulator and limit package, including comparison maps and the topology in which convergence is asserted.

The tuple is a type system for claims. A gluing amplitude can exist before it is shown to be a bounded map; a bounded positive map can exist before it is shown to be completely positive; a CP map at each cutoff can exist before it is shown to converge on a common completion.

Complete positivity is used only after the relevant operator systems or C*-completions and their matrix orderings have been declared. On a merely algebraic unbounded $*$-domain, the phrase *completely positive* requires an explicit matrix-positive cone and continuity/domain statement.

2.1 Boundary nets and gravitational localization

When localization is meaningful, each boundary Σ has a family of regions $U \subset \Sigma$ and a net $U \mapsto \mathcal{A}_\Sigma(U)$. At minimum, isotony $U \subset V \Rightarrow \mathcal{A}(U) \subset \mathcal{A}(V)$ is required. If a boundary causal structure is available, locality takes the form

$$[\mathcal{A}_\Sigma(U), \mathcal{A}_\Sigma(V)] = 0 \quad \text{for spacelike separated } U, V. \tag{3}$$

Gravity complicates (3): gauge-invariant observables may require dressing to an asymptotic or relational anchor and may carry edge data. Thus the local net is a physical claim, not decorative notation.

2.2 States and word Gram matrices

For a unital C*-algebra a state is a positive normalized functional. In field theory one often begins with a dense unbounded $*$-algebra, so domains must be printed. We use

$$\omega_\Sigma(a^*a) \geq 0, \qquad \omega_\Sigma(1) = 1, \quad a \in \mathcal{A}^0_\Sigma, \tag{4}$$

where $\mathcal{A}^0_\Sigma$ is a common test algebra. For words $a_1, \ldots, a_n$, positivity is equivalent to

$$G^\omega_{ij} = \omega(a_i^* a_j) \succeq 0. \tag{5}$$

This is the noncommutative analogue of Hankel positivity and contains ordered information absent from a commuting moment family.

3 GNS reconstruction and its exact scope

Theorem 3.1 (Algebraic GNS reconstruction)**.** *Let $\mathcal{A}^0$ be a unital $*$-algebra and let ω satisfy (4). Define*

$$N_\omega = \{a \in \mathcal{A}^0 : \omega(a^*a) = 0\}. \tag{6}$$

Then N_ω is a left ideal, $\mathcal{A}^0/N_\omega$ is a pre-Hilbert space with

$$\langle [a], [b] \rangle_\omega = \omega(a^*b), \tag{7}$$

and left multiplication gives a cyclic algebraic $*$-representation by generally unbounded operators on the common invariant domain $\mathcal{A}^0/N_\omega$. No boundedness or closability is asserted.

Proof. Cauchy–Schwarz follows by applying positivity to $\omega((a+\lambda b)^*(a+\lambda b))$ and optimizing over $\lambda \in \mathbb{C}$. If $a \in N_\omega$, Cauchy–Schwarz gives $\omega(a^*b) = 0$ for every b. Taking $b = x^*xa$ yields $\omega((xa)^*(xa)) = \omega(a^*x^*xa) = 0$, so $xa \in N_\omega$. This proves the left-ideal property and makes (7) and left multiplication well defined. The resulting algebraic representation has cyclic vector [1] [9, 20]. □

Remark 3.2 (Left-ideal warning). The GNS null space is generally a left ideal, not a two-sided $*$-ideal. Therefore $\mathcal{A}^0/N_\omega$ is naturally a representation space, not automatically a quotient algebra. A physical observable algebra $\mathcal{A}/N_\omega$ requires extra two-sided and involutive stability.

Nonclaim 3.3. GNS reconstruction proves neither locality, reflection positivity, Lorentzian unitarity, covariance, closability of the represented operators, a semiclassical metric, nor uniqueness of completion. It constructs the minimal cyclic algebraic representation of the supplied functional.

4 Physical reduction and quotient descent

Let $\mathcal{A}_{\text{test}}$ be the algebra before reduction. Three different kernels can appear: the GNS null left ideal N_ω; a constraint complex, such as a BRST differential s with $s^2 = 0$; and a two-sided $*$-ideal I_{phys} through which states and operations factor. The kernel of restriction to a small observer algebra is a fourth, epistemic object.

Definition 4.1 (Admissible physical reduction)**.** An admissible physical reduction is a specified construction

$$(\mathcal{A}_{\text{test}}, \omega, \Theta, \mathcal{S}, \mathcal{D}) \longrightarrow (\mathcal{A}_{\text{phys}}, \omega_{\text{phys}}, \Theta_{\text{phys}}, \mathcal{S}_{\text{phys}}, \mathcal{D}_{\text{phys}}) \tag{8}$$

such that every object on the right is well defined and the state remains positive. The branch may be BRST cohomology, Dirac reduction, positive averaging, a two-sided null quotient, or a declared composition.

These branches are not interchangeable. In a cohomological branch the state must annihilate exact representatives and induce a positive functional on the chosen cohomology; that construction need not be a C*-ideal quotient. The two quotient propositions below apply only to the stated closed-ideal branch.

Proposition 4.2 (Positive state descent)**.** *Let $\mathcal{A}$ be a unital C*-algebra, $I \triangleleft \mathcal{A}$ a closed two-sided $*$-ideal, and $q : \mathcal{A} \to \mathcal{A}/I$ the quotient map. A state ω factors as $\omega = \bar{\omega} \circ q$ if and only if $\omega(I) = 0$. The factor $\bar{\omega}$ is a state.*

Proof. Factorization of a linear functional through a quotient is equivalent to annihilating the kernel. If $x \in (\mathcal{A}/I)_+$, the C*-quotient has a positive lift $a \in \mathcal{A}_+$ with $q(a) = x$. Then $\bar{\omega}(x) = \omega(a) \geq 0$, independently of the lift. Normalization is inherited. □

Proposition 4.3 (CP quotient descent)**.** *Let $I \triangleleft \mathcal{A}$ and $J \triangleleft \mathcal{B}$ be closed two-sided $*$-ideals of unital C*-algebras. Let $\Phi : \mathcal{A} \to \mathcal{B}$ be completely positive and suppose $\Phi(I) \subset J$. Then there is a unique completely positive map*

$$\bar{\Phi} : \mathcal{A}/I \longrightarrow \mathcal{B}/J, \qquad \bar{\Phi}(q_A(a)) = q_B(\Phi(a)). \tag{9}$$

Proof. The ideal condition makes (9) well defined. At every matrix level, a positive element of the quotient has a positive lift. Complete positivity of Φ, followed by the positive quotient map q_B, therefore gives positivity of every amplification of $\bar{\Phi}$. □

These are standard operator-algebra statements. Their role is diagnostic: a sewing map that fails ideal preservation does not act on the claimed physical quotient.

5 The physical completion space

Definition 5.1 (Physically admissible completion space)**.** Fix a sector and all declared data in (2). The physical completion space K_{phys} is the set of normalized positive states on one fixed algebra $\mathcal{A}_{\mathrm{phys}}$ that satisfy:

(i) the chosen physical reduction and its domain conditions;
(ii) reflection positivity when Euclidean reconstruction is claimed;
(iii) matrix-level CP sewing and its composition laws;
(iv) the declared dynamics or Ward identities;
(v) regulator compatibility and the claimed limiting topology;
(vi) any semiclassical and infrared conditions included in the sector.

GNS or other representation data may be attached to an element of K_{phys}, but only together with its distinguished state. Bare representation classes are not elements of K_{phys}, because restriction to an observer subalgebra is then not defined.

Depending on the problem, K_{phys} may be empty, a point, a compact convex set, or a moduli space. Compactness is not assumed universally. When $\mathcal{A}_{\mathrm{phys}}$ is a unital C*-algebra and the gates define weak-* closed conditions, the state space is weak-* compact and K_{phys} is compact as a closed subset. Domain conditions for unbounded algebras can spoil this simple argument.

Remark 5.2 (Emptiness is information)*.* If no state satisfies all gates, the candidate sector is falsified as typed. One should not repair emptiness by silently changing the algebra, reflection, or regulator. Such a change defines a new candidate.

6 Restriction fibres and relational completeness

Let $R \subset \mathcal{A}_{\mathrm{phys}}$ be a unital observer or relational subalgebra. Restriction defines

$$\mathrm{res}_R : K_{\mathrm{phys}} \longrightarrow \mathsf{St}(R), \qquad \rho \longmapsto \rho|_R. \tag{10}$$

Definition 6.1 (Indeterminacy relative to R)**.** The sector has gravitational indeterminacy relative to R if res_R is noninjective: there exist $\rho_1 \neq \rho_2$ in K_{phys} such that

$$\rho_1|_R = \rho_2|_R. \tag{11}$$

Definition 6.2 (Relational completeness)**.** The algebra R is relationally complete for K_{phys} if res_R is injective. A family $\{R_i\}_{i\in I}$ is jointly complete if

$$\rho \longmapsto (\rho|_{R_i})_{i\in I} \tag{12}$$

is injective on K_{phys}.

Proposition 6.3 (Observable witness)**.** *The restriction map* res_R *is noninjective if and only if there are* $\rho_1, \rho_2 \in K_{\mathrm{phys}}$ *agreeing on* R *and a self-adjoint* $a \in \mathcal{A}_{\mathrm{phys}}$ *such that* $\rho_1(a) \neq \rho_2(a)$*.*

Proof. If restriction is noninjective, choose distinct states in the same fibre. Distinct linear functionals differ on at least one element of the full algebra. Writing that element as the sum of its self-adjoint real part and i times its self-adjoint imaginary part shows that the states differ on at least one self-adjoint element. The converse is immediate. □

The proposition is elementary; it is not the physical theorem. A gravitational realization must establish that R represents a physically motivated measurement class, both states pass every gate defining K_{phys}, the witness survives constraints and regulator limits, and the difference is not a gauge relabelling.

6.1 Relation to relational observables

Relational observables in canonical gravity use dynamical degrees of freedom as clocks or rods [7, 19]. The restriction formulation does not solve their construction. It assumes that a physical relational subalgebra has been identified and asks whether it separates admissible states. Clock choice is therefore part of the observer specification, not a free index on a scalar partition function.

6.2 Epistemic and physical fibres

Suppose a fixed-clock commutative moment problem is indeterminate. There may be many positive measures reproducing the same moments. Three outcomes are possible:

(i) additional noncommuting observables separate all measures, so the ambiguity was epistemic;
(ii) physical constraints remove all but one completion;
(iii) several completions remain in K_{phys} and differ on a physical observable, giving GIP relative to the restricted algebra.

Only the third outcome supports a physical completion landscape.

7 Scalar positivity and the amplified sewing test

The following distinction prevents a scalar diagnostic from being mistaken for complete positivity. It uses standard operator-algebra mathematics and is stated explicitly because it changes what a finite acceptance test must measure.

Proposition 7.1 (Scalar Gram consequence of positivity)**.** *Let $\Phi : A \to B$ be positive between unital C*-algebras and let ρ be a positive functional on B. Then every scalar matrix $[\rho(\Phi(a_i^* a_j))]$ is positive.*

Proof. For $x = \sum_j c_j a_j$ its quadratic form is $\rho(\Phi(x^* x)) \geq 0$. Complete positivity is not needed. □

Even quantifying over every word family and every scalar target state cannot certify CP. Transposition $T : M_2 \to M_2$ is positive and passes all those tests, while its Choi matrix is the flip, with eigenvalues $-1, 1, 1, 1$. The omitted direction is an entangled ancillary test, not another scalar word.

Proposition 7.2 (Operator-valued Gram characterization)**.** *A linear map $\Phi : A \to B$ is CP if and only if $[\Phi(a_i^* a_j)] \geq 0$ in $M_n(B)$ for every finite family $a_1, \ldots, a_n \in A$.*

Proof. Necessity follows by amplifying a positive Gram matrix. Conversely take an arbitrary $X \geq 0$ in $M_n(A)$ and factor $X = Y^* Y$. Then $X_{ij} = \sum_k Y_{ki}^* Y_{kj}$. Each row of Y gives a tested positive operator-valued Gram, so their sum $[\Phi(X_{ij})]$ is positive. This is positivity of every amplification. □

The finite constructions below establish CP through Kraus or isometry formulas; their scalar OS and state Grams are additional tests, not substitutes for this one.

8 Typed positivity and sewing gates

8.1 Scalar moments and noncommutative Gram matrices

Let $\mathcal{A}$ be a unital $*$-algebra. A normalized functional $\omega : \mathcal{A} \to \mathbb{C}$ is positive when $\omega(a^*a) \geq 0$ for every $a \in \mathcal{A}$ and $\omega(1) = 1$. If $z = z^*$ is one distinguished observable, its scalar moment sequence is

$$m_n = \omega(z^n), \qquad n \geq 0. \tag{13}$$

Hankel positivity tests the restriction of ω to the commutative algebra generated by z. For a polynomial $p(x) = \sum_i c_i x^i$,

$$\sum_{i,j} \overline{c}_i c_j m_{i+j} = \omega\big(p(z)^* p(z)\big) \geq 0. \tag{14}$$

This test contains no information about an operator not generated by z.

The noncommutative state test asks instead that every finite word family $a_1, \ldots, a_n \in \mathcal{A}$ have a positive Gram matrix

$$G^{\mathrm{nc}}_{ij} = \omega(a_i^* a_j) \succeq 0. \tag{15}$$

For a linear functional this is equivalent to positivity, because

$$c^* G^{\mathrm{nc}} c = \omega\left(\left(\sum_i c_i a_i\right)^* \left(\sum_j c_j a_j\right)\right).$$

The content is not the algebraic identity but the size of the tested algebra. A scalar moment tower is a shadow of the noncommutative condition, not a substitute for it.

8.2 Reflection form

Fix an anti-linear, multiplicative, star-preserving involution

$$\Theta(\lambda a) = \overline{\lambda}\Theta(a), \quad \Theta(ab) = \Theta(a)\Theta(b), \quad \Theta(a^*) = \Theta(a)^*, \quad \Theta^2 = \mathrm{id}. \tag{16}$$

Let $\mathcal{A}_+ \subseteq \mathcal{A}$ be a declared half-algebra or half-domain. The reflection form is

$$\langle a, b\rangle_{\mathrm{OS}} = \omega\big(\Theta(a) b\big), \qquad a, b \in \mathcal{A}_+. \tag{17}$$

Reflection positivity means that every finite matrix $[\omega(\Theta(a_i)a_j)]$ is positive semidefinite. It depends simultaneously on ω, Θ, and $\mathcal{A}_+$. Positivity of $\omega(a^*a)$ does not identify $\Theta(a)$ with a^*, so it does not imply (17). Conversely, a reflection-positive form on a small half-algebra says nothing by itself about a process map.

8.3 Complete positivity, Choi matrices, and effects

For unital C^*-algebras $\mathcal{A}_C$ and $\mathcal{A}_G$, a linear map $E : \mathcal{A}_C \to \mathcal{A}_G$ is completely positive if

$$E_n = \mathrm{id}_{\mathrm{M}_n} \otimes E : \mathrm{M}_n(\mathcal{A}_C)^+ \longrightarrow \mathrm{M}_n(\mathcal{A}_G)^+ \tag{18}$$

for every n. In finite dimensions, $E : \mathrm{M}_m \to \mathrm{M}_n$ is CP exactly when its Choi matrix

$$C_E = \sum_{i,j=1}^{m} e_{ij} \otimes E(e_{ij}) \tag{19}$$

is positive semidefinite. The convention (19) fixes which tensor factor carries the input matrix units; transposed conventions have the same spectrum but should not be mixed inside a numerical checker.

If $K_\alpha : \mathcal{H}_G \to \mathcal{H}_C$ are finite Kraus operators, then

$$E(a) = \sum_\alpha K_\alpha^* a K_\alpha \tag{20}$$

is CP. It is unital precisely when $\sum_\alpha K_\alpha^* K_\alpha = 1_G$ and subunital when the sum is at most 1_G. For an outcome x, the effect is

$$e_x = E_x(1_C), \qquad 0 \le e_x \le 1_G. \tag{21}$$

A finite instrument is a family $\{E_x\}_x$ of CP outcome maps satisfying $\sum_x e_x = 1_G$. For a glued state ω_G, it yields

$$p(x \mid \omega_G) = \omega_G(e_x), \qquad \omega_{C|x}(a) = \frac{\omega_G(E_x(a))}{p(x \mid \omega_G)} \tag{22}$$

when $p(x \mid \omega_G) > 0$. Complete positivity, subunitality, and total normalization are three separate checks.

Proposition 8.1 (Finite Stinespring compression, STD)**.** *Let $V : \mathcal{H}_G \to \mathcal{H}_C \otimes \mathcal{H}_E$ be linear and let $\pi(a) = a \otimes 1_E$. Then $E(a) = V^* \pi(a) V$ is CP. It is unital if and only if $V^*V = 1_G$.*

Proof. For $[a_{ij}] \succeq 0$ and vectors $\xi_i \in \mathcal{H}_G$,

$$\sum_{i,j} \langle \xi_i, E(a_{ij}) \xi_j \rangle = \sum_{i,j} \langle V\xi_i, \pi(a_{ij}) V \xi_j \rangle \ge 0.$$

The unit identity is $E(1) = V^*V$. □

8.4 Quotient descent and composition

Let $N_C \triangleleft \mathcal{A}_C$ and $N_G \triangleleft \mathcal{A}_G$ be closed two-sided star ideals. The formula

$$\overline{E}([a]) = [E(a)] \tag{23}$$

is representative independent exactly when

$$E(N_C) \subseteq N_G. \tag{24}$$

When E is CP and (24) holds, the induced map between C^*-quotients is CP. This statement applies to two-sided algebra ideals. A GNS left null space or a BRST cohomological quotient requires its own typing and cannot silently be inserted into (23).

For cuts $g : G \to C$ and $h : C \to D$, the Heisenberg order is

$$E_{h \circ g} = E_g \circ E_h : \mathcal{A}_D \longrightarrow \mathcal{A}_G. \tag{25}$$

Compositions and tensor products of CP maps are CP. Outcome refinements compose by $E_{(q,s)} = E_{g,q} \circ E_{h,s}$, and coarse graining is summation. Coherence therefore has an exact finite meaning once a compatible family of objects and maps is actually declared. A single cut does not define an entire bordism category.

Table 1: The typed gate hierarchy. The final column is deliberately not discharged by the finite algebraic test.

Gate	Datum	Finite test	Additional physical obligation
Scalar moment	$m_n = \omega(z^n)$	Hankel/localizing matrices	identify a physical observable and its domain
Noncommutative state	$\omega(a_i^* a_j)$	mixed Gram matrices	source the boundary algebra and state
Reflection	$(\Theta, \mathcal{A}_+)$	OS Gram matrix	justify geometric reflection and half-domain
Sewing	$E_g : \mathcal{A}_C \to \mathcal{A}_G$	Choi or Kraus test	derive E_g from a gravitational source
Instrument	$\{E_{g,x}\}$	effects sum to the unit	identify operational outcomes and records
Physical quotient	(N_C, N_G)	$E_g(N_C) \subset N_G$	prove nullness or constraint meaning
Coherence	compatible cut family	typed composition/tensor laws	control mapping-class, causal, and regulator limits

9 Exact finite separations between the gates

This section records exact counterexamples in the smallest convenient algebras. Their purpose is not novelty but error control: every omitted gate has a concrete failure mode. Together they prove that the arrows in Fig. 1 are a work program rather than a chain of automatic implications.

Theorem 9.1 (Scalar moment positivity does not imply noncommutative positivity)**.** *There is a normalized Hermitian functional on* $\mathrm{M}_2(\mathbb{C})$ *whose restriction to a projection has a positive determinate moment sequence, while the functional is not positive on* $\mathrm{M}_2(\mathbb{C})$*.*

Proof. Let $z = e_{22}$ and define $\varphi(a) = \mathrm{Tr}(Ra)$ with

$$R = \begin{pmatrix} \frac{1}{2} & 1 \\ 1 & \frac{1}{2} \end{pmatrix}. \tag{26}$$

Then $\varphi(1) = 1$ and the restriction to $C^*(z)$ is

$$\varphi(f(z)) = \tfrac{1}{2} f(0) + \tfrac{1}{2} f(1).$$

Consequently $m_0 = 1$ and $m_n = 1/2$ for $n \geq 1$, and every polynomial square has $\varphi(p(z)^* p(z)) = (|p(0)|^2 + |p(1)|^2)/2 \geq 0$. The representing measure is the two-atom measure $(\delta_0 + \delta_1)/2$. It is unique: any positive representing measure obeys $\int (x^2 - x)^2 \, d\mu = m_4 - 2m_3 + m_2 = 0$, hence is supported on $\{0, 1\}$, and m_1 fixes the two weights. However, R has eigenvalues $3/2$ and $-1/2$. For the rank-one projector $p_- = \frac{1}{2}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, $\varphi(p_-^* p_-) = \varphi(p_-) = -1/2$. The singleton noncommutative Gram matrix is therefore negative. □

Theorem 9.2 (A positive state need not be reflection positive)**.** *There is a faithful state* τ *on* $\mathrm{M}_2(\mathbb{C})$ *and an anti-linear multiplicative star-preserving involution* Θ *for which the OS form is negative on the full declared half-algebra.*

Proof. Take $\tau(a) = \mathrm{Tr}(a)/2$, $\Theta(a) = \overline{a}$, and declare $\mathcal{A}_+ = \mathrm{M}_2(\mathbb{C})$. Entrywise conjugation has all properties in (16). For

$$j = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

one has $\Theta(j) = j$ and $j^2 = -1$. Hence $\langle j, j\rangle_{\mathrm{OS}} = \tau(\Theta(j)j) = -1$, although τ is a faithful positive state and all matrices $[\tau(a_i^* a_j)]$ are positive. □

Theorem 9.3 (Reflection-compatible positivity does not imply complete positivity)**.** *There is a unital positive map that preserves a reflection-positive half-algebra and commutes with reflection, but whose Choi matrix has a negative eigenvalue.*

Proof. Again use $\tau = \mathrm{Tr}/2$ and $\Theta(a) = \overline{a}$, but choose the diagonal half-algebra $\mathcal{A}_+ = \{\mathrm{diag}(z_1, z_2)\}$. Then $\tau(\Theta(a)a) = (|z_1|^2 + |z_2|^2)/2 \geq 0$. Let $T(a) = a^{\mathsf{T}}$. Transposition is unital, positive, trace preserving, maps $\mathcal{A}_+$ to itself, and commutes with Θ. Its Choi matrix is the flip

$$C_T = \sum_{i,j=1}^{2} e_{ij} \otimes e_{ji}. \tag{27}$$

The flip has eigenvalue $+1$ on the three-dimensional symmetric subspace and -1 on the antisymmetric vector $(|01\rangle - |10\rangle)/\sqrt{2}$. Thus T is not CP. □

Proposition 9.4 (Complete positivity does not imply reflection covariance)**.** *Let $U = \mathrm{diag}(1, i)$ and $E(a) = U^* a U$ on $\mathrm{M}_2(\mathbb{C})$. Then E is a unital CP star automorphism, but it does not commute with $\Theta(a) = \overline{a}$.*

Proof. For e_{12}, $E(e_{12}) = ie_{12}$. Therefore $\Theta E(e_{12}) = -ie_{12}$ while $E\Theta(e_{12}) = ie_{12}$. □

Proposition 9.5 (Complete positivity does not imply normalization)**.** *For $K = 2\,1$ the map $E(a) = K^* a K = 4a$ is CP but $E(1) = 4\,1$. More generally, a Kraus representation proves CP independently of the effect inequality $\sum_\alpha K_\alpha^* K_\alpha \leq 1$.*

Proposition 9.6 (A unital CP map need not descend through a selected ideal)**.** *Let $\mathcal{A}_C = \mathcal{A}_G = \mathbb{C} \oplus \mathbb{C}$ and $E(a, b) = (b, b)$. The map is a unital star homomorphism and hence UCP. If $N_C = N_G = 0 \oplus \mathbb{C}$, then $E(0, b) = (b, b) \notin N_G$ for $b \neq 0$. Consequently (23) is not well defined.*

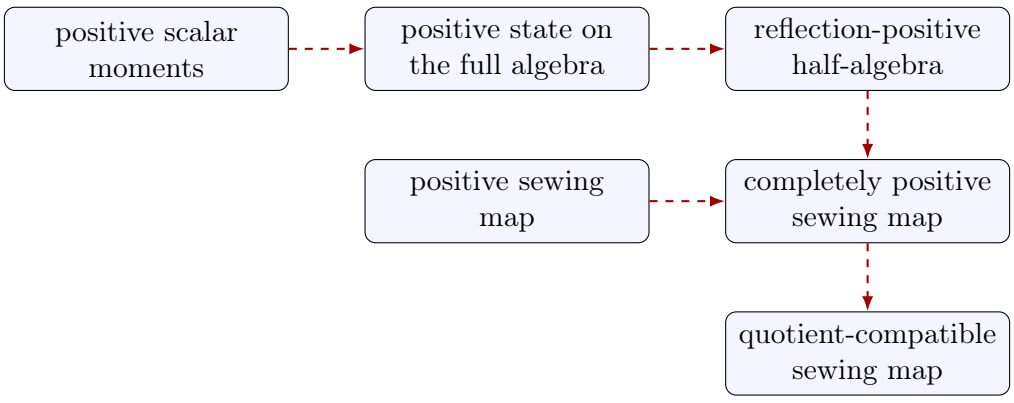


Figure 2: Exact nonimplications proved in this section. Every dashed red arrow denotes a desired upgrade that fails without an additional hypothesis.

The separations also clarify the word "hierarchy." Matrix CP is stronger than ordinary positivity for a fixed map, but reflection positivity and quotient descent are transverse requirements rather than simple levels of one order. A successful boundary process needs the intersection of these properties on compatible domains.

10 Doubled-boundary doubled finite operational model

We now construct a finite family in which the principal gates can coexist. The model is abstract operator algebra: the label "Doubled-boundary" records its intended use as a calibration for a future Teichmüller/Virasoro-TQFT compression, not an accomplished source identification.

10.1 Doubled channel datum

Let Q and P be finite glued and cut label sets, let $h = \mathbb{C}^d$, and set

$$K_x = \overline{h} \otimes h, \qquad \mathcal{A}_G = \bigoplus_{q \in Q} B(K_q), \qquad \mathcal{A}_C = \bigoplus_{p \in P} B(K_p). \tag{28}$$

The positive half-algebras are

$$\mathcal{A}_{G,+} = \bigoplus_{q} (1_{\overline{h}} \otimes B(h)), \qquad \mathcal{A}_{C,+} = \bigoplus_{p} (1_{\overline{h}} \otimes B(h)). \tag{29}$$

Choose a finite set of triples (p, α, q), coefficients $\kappa_{p\alpha|q} \geq 0$, unitaries $F_{p\alpha|q} \in U(h)$, and

$$s_q = \sum_{p,\alpha} \kappa_{p\alpha|q} \leq 1. \tag{30}$$

Define

$$L_{p\alpha q} = \sqrt{\kappa_{p\alpha|q}} (\overline{F}_{p\alpha|q} \otimes F_{p\alpha|q}) : K_q \to K_p \tag{31}$$

and the observable pullback

$$E_g(a)_q = \sum_{p,\alpha} L^*_{p\alpha q} a_p L_{p\alpha q}. \tag{32}$$

For a recorded outcome p, retain only the terms with that value of p. Complete the instrument by a failure algebra isomorphic blockwise to $\mathcal{A}_G$ and

$$E_{g,\varnothing}(b)_q = (1 - s_q) b_q. \tag{33}$$

The failure outcome is not cosmetic: without it a subunital cut would be silently renormalized and the outcome probabilities would not sum to one.

In the fixed basis of h, let the antiunitary swap be

$$J(\overline{\xi} \otimes \eta) = \overline{\eta} \otimes \xi, \qquad \Theta(a) = JaJ^{-1}. \tag{34}$$

Then $\Theta(1 \otimes A) = \overline{A} \otimes 1$ and $\Theta(\overline{F} \otimes F) = \overline{F} \otimes F$. The doubled form in (31) is selected precisely because it is fixed by the reflection.

Choose weights $w_q \geq 0$ with $\sum_q w_q = 1$, and define

$$\Omega_d = d^{-1/2} \sum_{i=1}^{d} \overline{e_i} \otimes e_i, \qquad \omega_G(a) = \sum_q w_q \langle \Omega_d, a_q \Omega_d \rangle. \tag{35}$$

Let $Q_* = \{q : w_q > 0\}$ and choose active cut labels $P_* \subseteq P$. The inactive whole-block ideals are

$$N_G = \bigoplus_{q \notin Q_*} B(K_q), \qquad N_C = \bigoplus_{p \notin P_*} B(K_p). \tag{36}$$

Assume the active-to-inactive exclusion

$$\kappa_{p\alpha|q} = 0 \quad \text{whenever } q \in Q_*,\ p \notin P_*. \tag{37}$$

These are genuine two-sided C^*-ideals, but they model deletion of entire inactive blocks only. They are not generic Ward, Dirac, or BRST ideals.

10.2 Finite operational theorem

Theorem 10.1 (Doubled finite operational sector, DERIVED)**.** *The data (28)–(37) have the following properties.*

(i) E_g *and every outcome map are CP. The completed instrument is unital, with* $E_g(1_C)_q = s_q 1_{K_q}$ *and* $E_{g,\varnothing}(1)_q = (1 - s_q)1_{K_q}$.

(ii) On the center $Z_C \cong \mathbb{C}^P$,

$$E_g(z)_q = \sum_p K(p \mid q) z_p 1_{K_q}, \qquad K(p \mid q) = \sum_\alpha \kappa_{p\alpha|q}, \tag{38}$$

so the center evolves by a substochastic Markov kernel completed by failure mass.

(iii) $E_g(\mathcal{A}_{C,+}) \subseteq \mathcal{A}_{G,+}$ *and* $E_g \Theta_C = \Theta_G E_g$*, outcome by outcome.*

(iv) The OS form on $\mathcal{A}_{G,+}$ *is*

$$\langle a, b \rangle_{\mathrm{OS}} = \sum_q \frac{w_q}{d} \operatorname{Tr}(A_q^* B_q), \qquad a_q = 1 \otimes A_q,\ b_q = 1 \otimes B_q. \tag{39}$$

Its radical is $N_G \cap \mathcal{A}_{G,+}$.

(v) The represented GNS kernel is N_G*. Condition (37) gives* $E_g(N_C) \subseteq N_G$*, hence a descended CP map* $\overline{E}_g : \mathcal{A}_C/N_C \to \mathcal{A}_G/N_G$.

(vi) If $u_x(t)$ *are one-particle unitary groups satisfying*

$$u_p(t) F_{p\alpha|q} = e^{i c_{p\alpha q} t} F_{p\alpha|q} u_q(t), \tag{40}$$

then the doubled dynamics $W_x(t) = \overline{u_x(t)} \otimes u_x(t)$ *commutes with the cut map.*

(vii) Successive cuts compose in the Heisenberg order $E_{G \to D} = E_{G \to C} \circ E_{C \to D}$*, and products of doubled Kraus operators are again doubled.*

Proof. The Kraus formula gives CP at every matrix level. Since $L_{p\alpha q}^* L_{p\alpha q} = \kappa_{p\alpha|q} 1$, summing the effects and adding (33) proves normalization. Central elements act as scalars in each cut block, giving (38). For $a_p = 1 \otimes A_p$,

$$L_{p\alpha q}^* a_p L_{p\alpha q} = \kappa_{p\alpha|q} (1 \otimes F_{p\alpha|q}^* A_p F_{p\alpha|q}),$$

which proves half-algebra preservation. The swap fixes every doubled Kraus block, so reflection covariance follows term by term.

The identity

$$\langle \Omega_d, (\overline{A} \otimes B) \Omega_d \rangle = \frac{1}{d} \operatorname{Tr}(A^* B) \tag{41}$$

gives (39). Strictness on every $w_q > 0$ block identifies the OS radical. On each active simple matrix block the represented GNS kernel is zero; inactive blocks are annihilated, so the total represented kernel is N_G. If a is supported only on an inactive cut block, (37) makes every active glued component of $E_g(a)$ vanish. This proves ideal preservation and CP descent.

Equation (40) yields $W_p(t) L_{p\alpha q} = L_{p\alpha q} W_q(t)$ because the ordinary and conjugate phases cancel. Finally, if a second cut has Kraus operators $M_{r\beta p}$, substitution gives

$$E_1(E_2(a)) = \sum_{p,\alpha,r,\beta} (M_{r\beta p} L_{p\alpha q})^* a_r (M_{r\beta p} L_{p\alpha q}).$$

Operator multiplication is associative, and $(\overline{G} \otimes G)(\overline{F} \otimes F) = \overline{GF} \otimes GF$. □

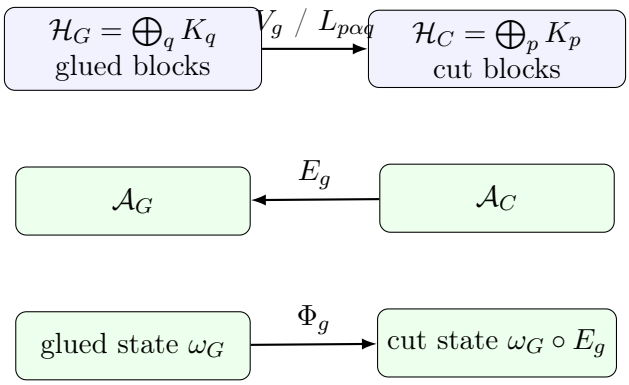


Figure 3: Variance and normalization in the Doubled-boundary finite model. Hilbert-space arrows point from glued to cut, observable maps point back, and state duals point forward. Subunital outcome effects are completed separately by the failure effect $1 - s_q$.

10.3 Interpretive ceiling

The theorem shows that finite matrix cones, CP cuts, reflection covariance, OS positivity, block-ideal descent, compatible finite dynamics, and associative composition are jointly consistent. It does not identify p or q with a principal-series momentum, F with a fusion kernel, or the Markov center with a gravitational edge observable. The finite labels, weights, unitaries, and dynamics were selected. A physical three-dimensional result must instead obtain them by a controlled compression of the positive Ponsot–Teschner/CEZ data and must prove that the compression errors remain compatible under refinement.

11 Two-packet instance and deterministic audit

Take $h = \mathbb{C}^2$, $K = \overline{h} \otimes h \cong \mathbb{C}^4$, one glued block, and two cut blocks. Let $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $U = X \otimes X$, and

$$K_0 = \frac{1}{\sqrt{2}} 1_4, \qquad K_1 = \frac{1}{\sqrt{2}} U, \qquad E_g(a_0 \oplus a_1) = \frac{1}{2}(a_0 + U^* a_1 U). \tag{42}$$

The effects are both $1_4/2$, so no failure outcome is needed. Choose

$$\Omega_\lambda = \sqrt{\lambda}|00\rangle + \sqrt{1-\lambda}|11\rangle, \qquad 0 < \lambda < 1, \quad \lambda \neq \tfrac{1}{2}. \tag{43}$$

The two outcomes have probability $1/2$. After identifying their matrix fibers, the posterior densities are $\rho_0 = |\Omega_\lambda\rangle\langle\Omega_\lambda|$ and $\rho_1 = U\rho_0 U^*$. Their pure-state overlap and trace distance are

$$|\langle \Omega_\lambda, U\Omega_\lambda\rangle| = 2\sqrt{\lambda(1-\lambda)}, \qquad \frac{1}{2}\left\|\rho_0 - \rho_1\right\|_1 = |2\lambda - 1|. \tag{44}$$

As states on the full direct-sum outcome algebra, however, the conditioned states have orthogonal central support and trace distance one. The distinction between common-fiber distance and recorded-outcome distance is operationally important.

Let $D = \mathrm{diag}(\lambda, 1-\lambda)$. For $A, B \in \mathrm{M}_2(\mathbb{C})$, the swap reflection gives

$$\omega_\lambda\big(\Theta(1 \otimes A)(1 \otimes B)\big) = \mathrm{Tr}(D^{1/2} A^* D^{1/2} B). \tag{45}$$

In the matrix-unit basis the four OS eigenvalues are

$$\lambda, \quad \sqrt{\lambda(1-\lambda)}, \quad \sqrt{\lambda(1-\lambda)}, \quad 1-\lambda, \tag{46}$$

Table 2: Rerun of the Doubled-boundary two-packet checker. Analytic identities are the proof; the floating-point values are regression evidence.

Quantity	Rerun value
$\|V^*V - 1\|$	2.22×10^{-16}
instrument normalization residual	2.22×10^{-16}
minimum outcome Choi eigenvalue	-5.55×10^{-17}
reflection-covariance residual (seeded sample)	0
smallest OS Gram eigenvalue	0.3000000000000001
OS spectrum-formula residual	1.24×10^{-16}
identified-fiber posterior distance	0.4
recorded-outcome posterior distance	1
generator spectrum	$0, 1.3, 1.3, 2.6$
overall	Pass

so the form is positive definite for $0 < \lambda < 1$. If $h = \mathrm{diag}(0, \Delta)$, the semigroup $T_t(A) = e^{-th}Ae^{-th}$ is self-adjoint and contractive in this OS inner product; its positive generator $L_h + R_h$ has spectrum

$$0, \quad \Delta, \quad \Delta, \quad 2\Delta. \tag{47}$$

This is a finite reconstruction calibration, not physical Lorentzian time.

The companion script was rerun for $\lambda = 0.7$, $\Delta = 1.3$, and tolerance 10^{-12}. Selected results are shown in Table 2. The Choi minimum of -5.55×10^{-17} is consistent with roundoff and is far above the negative tolerance threshold. Invalid controls deliberately detect superunital Kraus weights, a dropped outcome, a singular OS state, and an obsolete reflection convention.

The checker exhausts the declared two-outcome Choi matrices, the OS basis, posterior formulae, and generator spectrum. Reflection is checked on a seeded sample in addition to the analytic proof. It does not numerically instantiate every general ideal, dynamics, failure-posterior, or multi-cut case in Section 10.

12 Finite Weyl charge, dressing, and matched sewing

The second calibration family has a different purpose. It tests whether charge sectors, an invariant subalgebra, a neutralizing anchor, a retained edge center, and a matched-sector cut can coexist with CP and exact quotient descent. Slot names are mnemonic only. In particular, a slot called g_+ or g_- is not a graviton helicity, and a slot called ϕ is not a continuum field.

12.1 Charge: discrete Weyl algebra and charge physicalization

Fix an odd prime d, let $\zeta = e^{2\pi i/d}$, and let $h = \mathbb{C}^d$. With $X|n\rangle = |n+1\rangle$ and $Z|n\rangle = \zeta^n|n\rangle$, define for $u = (a, b) \in \mathbb{F}_d^2$

$$W(u) = \zeta^{ab/2}X^aZ^b, \qquad \sigma(u, v) = ab' - ba'. \tag{48}$$

All exponents are evaluated in $\mathbb{F}_d$. Direct multiplication gives

$$W(u)W(v) = \zeta^{-\sigma(u,v)/2}W(u+v), \qquad W(u)^* = W(-u), \qquad \mathrm{Tr}(W(u)^*W(v)) = d\,\delta_{u,v}. \tag{49}$$

The finite Weyl relation is not the continuum CCR. Indeed, no finite-dimensional matrices satisfy $[Q, P] = i\hbar 1$ because the traces of the two sides disagree.

For each side $r \in \{L, R\}$ take four slots $S = \{g_+, g_-, \phi, a\}$ and

$$\mathcal{H}_r^{\mathrm{kin}} = \bigotimes_{s \in S} h_{r,s}, \qquad \mathcal{A}_r^{\mathrm{kin}} = B(\mathcal{H}_r^{\mathrm{kin}}) \cong \mathrm{M}_{d^4}(\mathbb{C}). \tag{50}$$

Define the finite charge representation

$$U_r(t) = Z_{r,g_+}^t Z_{r,g_-}^t Z_{r,\phi}^t Z_{r,a}^t, \qquad t \in \mathbb{F}_d, \tag{51}$$

and Fourier projectors

$$P_{r,q} = \frac{1}{d} \sum_{t \in \mathbb{F}_d} \zeta^{-qt} U_r(t). \tag{52}$$

Character orthogonality implies

$$P_{r,q} P_{r,q'} = \delta_{q,q'} P_{r,q}, \qquad \sum_q P_{r,q} = 1, \qquad \operatorname{rank} P_{r,q} = d^3. \tag{53}$$

The finite-group average

$$\mathcal{F}_r(A) = \frac{1}{d} \sum_{t \in \mathbb{F}_d} U_r(t) A U_r(t)^* \tag{54}$$

is a UCP conditional expectation onto

$$\mathcal{A}_r^{\mathrm{inv}} = \bigoplus_{q \in \mathbb{F}_d} B(P_{r,q} \mathcal{H}_r^{\mathrm{kin}}), \qquad Z_r^{\mathrm{edge}} = \operatorname{span}\{P_{r,q}\}. \tag{55}$$

The proof is finite: each conjugation is a UCP star automorphism, their average is UCP, group translation proves idempotence, and the spectral decomposition identifies the range. For any density matrix ρ, the functional $\omega^{\mathrm{inv}}(A) = \operatorname{Tr}(\rho\, \mathcal{F}_r(A))$ is positive on the entire mixed four-slot algebra. This mixed state cone is stronger than four separate slot tests.

12.2 Dressing: neutral anchor dressing

Let $u = (u_{g_+}, u_{g_-}, u_\phi)$ label the three internal slots, with $u_s = (a_s, b_s)$, and put

$$c_{\mathrm{int}}(u) = a_{g_+} + a_{g_-} + a_\phi \in \mathbb{F}_d. \tag{56}$$

The neutral dressed Weyl generator is

$$D_{r,0}(u) = W_r^{\mathrm{int}}(u) X_{r,a}^{-c_{\mathrm{int}}(u)}. \tag{57}$$

Because anchor powers commute among themselves and act on a distinct tensor factor, the internal Weyl multiplier is unchanged:

$$D_{r,0}(u) D_{r,0}(v) = \zeta^{-\Sigma_{\mathrm{int}}(u,v)/2} D_{r,0}(u+v), \qquad D_{r,0}(u)^* = D_{r,0}(-u). \tag{58}$$

Thus $W_r^{\mathrm{int}}(u) \mapsto D_{r,0}(u)$ extends to an injective unital star homomorphism

$$\iota_{r,0} : \mathrm{M}_{d^3}(\mathbb{C}) \hookrightarrow \mathcal{A}_r^{\mathrm{kin}}. \tag{59}$$

The internal and anchor charge phases cancel, so $\operatorname{Ad} U_r(t)(D_{r,0}(u)) = D_{r,0}(u)$. Consequently the dressed algebra lies in $\mathcal{A}_r^{\mathrm{inv}}$ and acts faithfully in every charge block.

Changing an anchor origin by $s \in \mathbb{F}_d$ gives

$$\iota_{r,s} = \operatorname{Ad}(Z_{r,a}^s) \circ \iota_{r,0}, \qquad C_r(s_2, s_1) = Z_{r,a}^{s_2 - s_1}, \tag{60}$$

with the exact cocycle law $C_r(s_3,s_2)C_r(s_2,s_1)=C_r(s_3,s_1)$. Computational-basis conjugation satisfies $\Theta_r(D_{r,0}(u))=D_{r,0}(\kappa u)$, where $\kappa(a,b)=(a,-b)$. This is reflection covariance, not OS positivity.

A useful positive control is the bounded slot quadratic

$$T_{\mathrm{slot},r}=(1-Y_r)^*(1-Y_r)+(1-Z_r^{\mathrm{dress}})^*(1-Z_r^{\mathrm{dress}}), \tag{61}$$

where $Y_r=X_{r,\phi}X_{r,a}^{-1}$ and $Z_r^{\mathrm{dress}}=Z_{r,\phi}$. It is positive, charge invariant, and anchor covariant. It has no spacetime index, conservation law, or Ward identity and is not a stress tensor.

12.3 Cut: matched-charge cut and quotient

Define the cut and glued Hilbert spaces

$$\mathcal{H}_C=\bigoplus_{q_L,q_R\in\mathbb{F}_d}\mathcal{H}_L(q_L)\otimes\mathcal{H}_R(q_R), \tag{62}$$

$$\mathcal{H}_G=\bigoplus_{q\in\mathbb{F}_d}\mathcal{H}_L(q)\otimes\mathcal{H}_R(-q), \tag{63}$$

and let $\mathcal{A}_C$ and $\mathcal{A}_G$ be the corresponding block-diagonal direct sums of full matrix algebras. Let $V_g:\mathcal{H}_G\to\mathcal{H}_C$ be the canonical inclusion into the matched subspace. If Π_q^C projects onto the cut block $(q,-q)$, set

$$E_g(a)=V_g^*aV_g,\qquad E_{g,q}(a)=V_g^*\Pi_q^Ca\Pi_q^CV_g. \tag{64}$$

Theorem 12.1 (Matched-charge finite process package, DERIVED)**.** *The map $E_g:\mathcal{A}_C\to\mathcal{A}_G$ is UCP and the $E_{g,q}$ form a normalized CP instrument with effects e_q^G. On the cut center,*

$$E_g(\Pi^C_{q_L,q_R})=\delta_{q_R,-q_L}e^G_{q_L}. \tag{65}$$

The unmatched block algebra

$$I_{\mathrm{unmatch}}=\bigoplus_{q_L+q_R\neq0}B(\mathcal{H}_L(q_L)\otimes\mathcal{H}_R(q_R)) \tag{66}$$

is a two-sided star ideal, $\ker E_g=I_{\mathrm{unmatch}}$*, and* $\mathcal{A}_C/I_{\mathrm{unmatch}}\cong\mathcal{A}_G$*. On the declared block algebras, E_g is the associated unital star quotient homomorphism. It preserves the dressed Weyl relations, edge center, slot quadratics, and computational reflection. A compatible family of such inclusions composes in the order* (25) *and has exact outcome refinement and tensor-product coherence.*

Proof. Compression by the isometry V_g proves UCP. Replacing V_g by $\Pi_q^CV_g$ proves CP of each outcome, and orthogonality of the matched block projectors gives $\sum_q E_{g,q}=E_g$ and $\sum_q E_{g,q}(1_C)=1_G$. The inclusion has range only in $(q,-q)$ blocks, yielding (65). Because $\mathcal{A}_C$ is explicitly block diagonal, restriction deletes exactly the unmatched full matrix blocks and preserves products and stars on every matched block. This proves the kernel and quotient statements. All dressed generators and center projectors are block diagonal, so their relations survive restriction. The inclusion is real in the computational basis, which gives $\Theta_GE_g=E_g\Theta_C$. Composition follows from $(V_hV_g)^*a(V_hV_g)=V_g^*(V_h^*aV_h)V_g$; outcome and tensor statements follow by the same calculation. □

For a glued state ρ, the outcome probabilities and normalized cut posteriors are

$$p(q\mid\rho)=\rho(e_q^G),\qquad\Phi_{g,q}(\rho)(a)=\frac{\rho(E_{g,q}(a))}{p(q\mid\rho)}. \tag{67}$$

The reverse observable inclusion $J_g(b) = V_g b V_g^*$ is an injective nonunital star homomorphism with $J_g(1_G) = \Pi_{\mathrm{match}}$. A cut-to-glued conditioned state rule

$$\Gamma_g(\sigma)(b) = \frac{\sigma(J_g(b))}{\sigma(\Pi_{\mathrm{match}})} \tag{68}$$

is nonlinear on the full state space. Neither J_g nor Γ_g is the state dual $\Phi_g = \rho \circ E_g$.

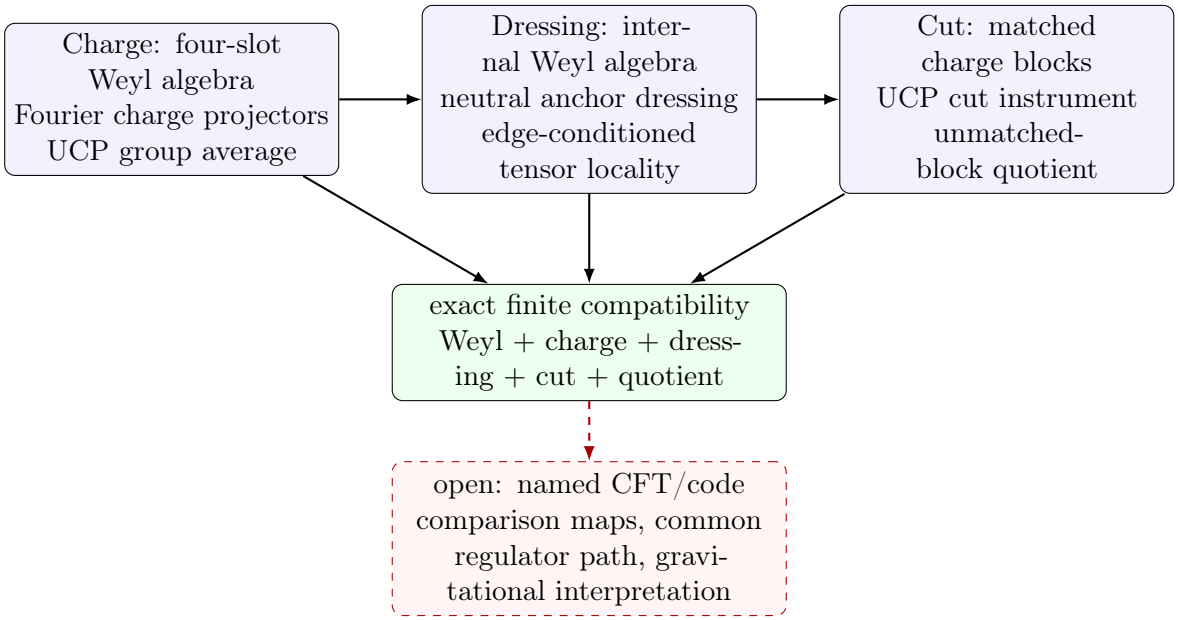


Figure 4: The Weyl/charge/cut logical flow. Exact finite compatibility terminates before the physical bridge.

12.4 Rerun evidence and verification ceiling

At $d = 3$, the companion checker exhausts the one-slot Weyl labels and the charge projector identities. It samples a group-average input, a positive averaged output, one dressed Weyl pair, and a cut state using a fixed seed. The rerun values in Table 3 all satisfy the declared tolerance 2×10^{-11}.

The checker detects three invalid controls: a naively truncated CCR, an undressed charged operator, and an incomplete instrument. It does not construct the full Choi matrix of the 6561-dimensional matched-charge map, nor is that needed for the theorem: the analytic compression proof establishes CP at every matrix level. The numerical run is a regression check on selected coordinates, not a substitute for that proof.

13 A deterministic acceptance protocol

The two models suggest a compact protocol that separates analytic obligations from finite implementation checks. Given finite algebras, outcome maps, reflections, and ideals, one first fixes map variance and bases. The Choi matrix is then constructed from (19) or CP is proved from an explicit Stinespring/Kraus representation. Effects are evaluated independently and summed to the identity. Reflection is checked as an anti-linear map, including multiplicativity and star compatibility, before its intertwining with each outcome map is tested. Quotient descent is verified by applying the map to a basis of the outgoing ideal and checking membership

Table 3: Selected rerun values for the $d = 3$ Weyl/charge/dressing/cut checker.

Quantity	Rerun value
single-mode Weyl product residual	2.59×10^{-15}
projector completeness residual	1.91×10^{-15}
projector ranks	$27, 27, 27$
sample group-average idempotence residual	5.05×10^{-14}
sample positive-output invariance residual	1.40×10^{-12}
dressing charge-invariance residual	6.86×10^{-15}
sample dressed-Weyl residual	2.37×10^{-14}
anchor-change residual	4.73×10^{-15}
T_{slot} minimum eigenvalue	1.2679491924
cut-space / glued-space dimensions	6561/2187
sample outcome probabilities	$0.2, 0.3, 0.5$
instrument probability residual	1.11×10^{-16}
successive-cut charge-set difference	0
overall	PASS

in the incoming ideal. Finally, sequential maps are compared in the Heisenberg order (25); outcome sums are checked before any conditioned renormalization.

Table 4: Acceptance protocol and the strength of each evidence type.

Check	Exact criterion	Finite implementation	What a pass does not prove
CP	$C_E \succeq 0$ or $E(a) = V^*\pi(a)V$	minimum Choi eigenvalue with tolerance	physical origin of E
Effect	$0 \le E_x(1) \le 1$	effect eigenvalues	total normalization
Instrument	$\sum_x E_x(1) = 1$	norm residual	interpretation of outcomes
Reflection type	(16)	basis and random-product residuals	geometric OS reflection
OS positivity	$[\omega(\Theta(a_i)a_j)] \succeq 0$	declared half-basis Gram spectrum	continuum reconstruction axioms
Ideal descent	$E(N_C) \subseteq N_G$	ideal-basis membership	physical nullness of the ideal
Composition	$E_{h\circ g} = E_g E_h$	matrix residual or Kraus multiplication	a complete cut category
Source bridge	comparison square and error bounds	not supplied here	continuum gravity

A tolerance must be paired with a scaling convention. A minimum Choi eigenvalue of -10^{-16} is harmless only when matrix norms are order one and the declared tolerance is much larger. Conversely, a sampled residual of zero does not establish an identity on an untested space. The correct report therefore labels tests as exhaustive, analytic, or seeded-sample. Both scripts used here print their scope and deliberately failing controls; that practice guards against a checker whose pass condition is vacuous.

14 The missing physical source bridge

Finite consistency is a necessary but weak result. The source bridge asks whether the finite data arise as controlled compressions of a specified gravitational construction and whether the compatibility gates survive a directed limit. This is not a matter of renaming the finite labels.

14.1 Three-dimensional source comparison

A Doubled-boundary promotion would begin with an actual principal-series Hilbert/form datum, Ponsot–Teschner or Virasoro-TQFT sewing kernels, and their positive row measure [4, 18]. For a regulator R, one would need packet embeddings or compressions

$$C_{X,R} : \mathcal{A}_X^{\mathrm{fin}}(R) \longrightarrow \mathcal{A}_X^{\mathrm{src}}(R) \tag{69}$$

and a source cut map $E_g^{\mathrm{src}}(R)$. The finite square has a meaningful defect only after all maps exist:

$$\varepsilon_{\mathrm{cut}}^{\mathrm{cb}}(R) = \left\| C_{G,R} E_g^{\mathrm{fin}} - E_g^{\mathrm{src}}(R) C_{C,R} \right\|_{\mathrm{cb}} . \tag{70}$$

Unital, state, center, reflection, presentation, composition, and dynamics defects must be bounded on one common domain. A polar correction or failure outcome may repair a finite compression, but its correction norm must also tend to zero. In addition, the directed family must preserve the matrix cones and states rather than merely converge entrywise. No such complete comparison family is constructed here; consequently $\varepsilon_{\mathrm{cut}}^{\mathrm{cb}}$ is presently undefined, not zero.

The distinction is substantive. Virasoro TQFT describes a precise fixed-topology on-shell sector and brings noncompact fusion and mapping-class structures that are absent from a generic finite matrix channel [4, 5]. Compact packet windows must control leakage, branch and chart conventions, discrete or vacuum contributions, and the convergence topology of sums of CP correspondences. The Doubled-boundary theorem provides the target type for such a comparison but proves none of those estimates.

14.2 Four-dimensional code comparison

For the Weyl/charge model, a physical bridge requires a named sequence of CFT or code algebras, not only an analogy with holographic codes. One would need maps

$$C_{C,\lambda} : \mathcal{A}_C^{\mathrm{fin}} \to \mathcal{A}_C^{\mathrm{code}}(\lambda), \qquad C_{G,\lambda} : \mathcal{A}_G^{\mathrm{fin}} \to \mathcal{A}_G^{\mathrm{code}}(\lambda), \tag{71}$$

together with a physical $E_g^{\mathrm{code}}(\lambda)$. The analogue of (70) must vanish along a declared large-N/cutoff path, while product, star, state, charge, dressing, localization, reflection, instrument, posterior, and coherence errors vanish compatibly.

Holographic codes show that finite isometries can capture important reconstruction and entropy patterns [10, 17]. They do not license identifying the present four slots with gravitons and matter. Moreover, actual gravitational dressing obstructs naive exact tensor-factor locality [8]. The exact commutativity of the left and right finite tensor factors is therefore a surrogate control, not evidence for locality of observables with shared asymptotic dressing. A physical bridge must quantify that discrepancy rather than hide it in slot names.

14.3 Constraint meaning and continuum limits

In both families the quotient ideals are elementary whole-block ideals. Their algebraic descent is exact, but their physical meaning is selected by hand. A gravitational quotient would normally

involve constraints, Ward identities, edge charges, gauge cohomology, or state-dependent null spaces. One must prove that the relevant null structure is stable under every sewing map and reflection, and that the quotient cone remains proper. It is not sufficient to observe that a chosen direct-sum block can be deleted.

Similarly, a directed limit needs compatible embeddings, uniformly controlled maps, states, and reflections, and a declared topology. Entrywise convergence of finite matrices does not automatically preserve complete positivity on an unbounded dense domain. Even after a positive continuum process is obtained, semiclassical Einstein dynamics and an infrared observable map remain additional obligations. This paper therefore stops at finite calibration by design.

15 Discussion

The central result is negative and constructive at once. Negatively, scalar moments, positive states, reflection forms, positive process maps, CP maps, normalized instruments, and physical quotients are not interchangeable. The elementary examples in Section 9 expose a distinct failure at each substitution. Constructively, the Doubled-boundary and Weyl/charge/cut families demonstrate that selected versions of all these interfaces can coexist without contradiction in finite matrix algebra.

The doubled-boundary model is strongest where reflection and operational normalization are concerned. The choice $\overline{F} \otimes F$ simultaneously preserves the half-algebra, fixes the maximally entangled vector, and cancels phases in both reflection and dynamics. The failure block turns subunital cutting into a genuine normalized instrument. The price is rigidity: these conveniences are engineered. A source kernel need not have exactly doubled unitary blocks, and a nonmaximally entangled state requires a separate OS proof, as the two-packet example illustrates.

The Weyl/charge model is strongest where edge centers, invariant reduction, dressing, and quotient matching are concerned. Finite-group averaging supplies a standard UCP conditional expectation, and neutral anchor dressing embeds a full internal matrix algebra faithfully into every retained charge sector. Matched-sector compression is not merely CP but a star quotient homomorphism on the declared block-diagonal algebra. The price is again physical distance: modulo-d charge is not a gravitational Gauss law, the anchor is not a gravitational line dressing, and T_{slot} is not a stress tensor.

The literature boundary is correspondingly narrow. Stinespring representations, Choi matrices, Kraus operations, instruments, OS reconstruction, conditional expectations, and operator-algebra quantum error correction are established structures [2, 3, 6, 15, 21]. The contribution of the present framework is not their invention. It is the explicit requirement that matrix CP, operational normalization, reflection, ideal descent, edge records, and composition be checked on one typed family, plus exact finite reference implementations and explicit stop lines. Whether this synthesis has gravitational content is decided only by the source bridges in Section 14.

The finite models can still be scientifically useful before that bridge exists. They serve as unit tests for proposed continuum formulae. A candidate compression that produces a sign-indefinite Choi matrix, loses a failure outcome, deletes an active edge block, reverses map variance, or violates reflection covariance fails immediately. They also distinguish a genuine obstruction from numerical instability: a transpose map has an exact Choi eigenvalue -1, whereas the two-packet Choi minimum of order 10^{-17} is roundoff around an analytic positive matrix.

16 Limitations and nonclaims

The main mathematical limitation is finite dimensionality. Matrix algebras avoid unbounded operator domains, type-III local structure, infrared sectors, anomalies, and the topology of regulator removal. The quotient ideals are whole direct-sum blocks, not cohomological or dynamical constraint ideals. The OS forms are finite Gram forms, not a verification of all Osterwalder–Schrader axioms. Dynamics in the Doubled-boundary model is a selected finite unitary or contraction semigroup, not a Hamiltonian constraint or physical time.

The numerical evidence is intentionally subordinate to the proofs. The Doubled-boundary checker exhausts its small Choi and OS matrices but samples some covariance identities. The $d = 3$ Weyl checker exhausts one-slot Weyl and projector identities but samples group-average and dressing inputs and does not build the full matched-charge Choi matrix. Passing results show that the implementations agree with selected analytic formulae at one finite regulator. They do not establish a convergence theorem.

No result here establishes a CEZ packet origin, a Moore–Seiberg coherent cut category, a physical AdS/CFT code, a graviton, a scalar QFT, a renormalized stress tensor, diffeomorphism constraints, anomaly cancellation, semiclassical Einstein equations, gravitational EFT recovery, a UV completion, or an empirical prediction. The title's "physical quotient" should therefore be read as a typed target instantiated by exact finite block quotients, not as a claim that the model's ideals have already been identified with the null ideal of quantum gravity.

17 Conclusion

A boundary quantum theory cannot be certified by one scalar notion of positivity. Noncommutative state positivity, reflection positivity, matrix-completely-positive sewing, effect normalization, quotient descent, and coherent composition answer different questions. Exact 2×2 counterexamples show that omitting any of these questions can change the answer from positive to negative.

The two finite families constructed here pass a substantial common subset of the resulting gate stack. They provide exact formulas, analytic proofs, deterministic checks, invalid controls, and a clear map variance. This is enough to establish the internal consistency and nontriviality of the selected operator-algebraic interfaces. It is not enough to establish quantum gravity. The decisive next result must construct one physical source family and prove that its comparison maps, CP sewing, reflection, constraints, edge records, and regulator limit close together. Until then, the finite models should be used as calibration standards and falsifiers, not as evidence for a continuum gravitational sector.

A Choi and instrument conventions

For $E : \mathrm{M}_m \to \mathrm{M}_n$ our Choi convention is (19). If $E(a) = K^* a K$ with $K : \mathbb{C}^n \to \mathbb{C}^m$, then vectorization gives

$$C_E = \mathrm{vec}(K)\,\mathrm{vec}(K)^* \tag{72}$$

up to the fixed vectorization order, and is therefore positive. A Kraus sum gives a sum of such rank-one matrices. The eigenvalues depend neither on a simultaneous basis change nor on which consistent Choi tensor convention is chosen.

In Heisenberg variance, $E_x(1)$ is an effect in the input/glued algebra. If ω_G is a state, the probability is $\omega_G(E_x(1))$. The normalized posterior is a state on the outcome algebra, not an operator in the glued algebra. For a direct sum of recorded outcomes, two conditioned states in

disjoint blocks have trace distance one even when their internal fiber states are close. Forgetting the center is a separate channel and should be declared before comparing fiber densities.

For successive cuts, if $K_\alpha : \mathcal{H}_G \to \mathcal{H}_C$ and $M_\beta : \mathcal{H}_C \to \mathcal{H}_D$, then

$$E_{G\to C}(E_{C\to D}(a)) = \sum_{\alpha,\beta} (M_\beta K_\alpha)^* a (M_\beta K_\alpha). \tag{73}$$

This formula fixes both the map order and the product order of Kraus operators.

B Expanded verification of the finite separations

For the scalar-moment example, every Hankel quadratic is explicitly

$$\sum_{i,j=0}^{N} \overline{c}_i c_j m_{i+j} = \frac{1}{2}\left|\sum_{i=0}^{N} c_i 0^i\right|^2 + \frac{1}{2}\left|\sum_{i=0}^{N} c_i\right|^2, \tag{74}$$

with the convention $0^0 = 1$. It is nonnegative at every order. The negative noncommutative direction p_- is invisible because $p_- \notin C^*(e_{22})$.

For transposition on M_2, the Choi matrix in the ordered basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ is

$$C_T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{75}$$

The central 2×2 block has eigenvalues ± 1. Thus a scalar positivity test or a test on the diagonal half-algebra cannot see the ancillary antisymmetric direction that violates CP.

For the quotient counterexample, the quotient map $q : \mathbb{C}\oplus\mathbb{C} \to \mathbb{C}$ associated with $N = 0\oplus\mathbb{C}$ keeps the first coordinate. If a descended map $\overline{E}$ existed, it would obey $\overline{E}(q(0,b)) = q(E(0,b))$. The left side is $\overline{E}(0) = 0$, while the right side is $q(b,b) = b$. This contradiction proves failure of representative independence.

C Doubled-boundary proof details

For the posterior formula in the maximally entangled FOS state, the doubled unitary fixes Ω_d:

$$(\overline{F} \otimes F)\Omega_d = \Omega_d. \tag{76}$$

Consequently the unnormalized outcome functional is

$$\Phi_{g,p}(\omega_G)(a_p) = \sum_{q,\alpha} w_q \kappa_{p\alpha|q} \langle \Omega_d, a_p \Omega_d \rangle, \tag{77}$$

and

$$p(p \mid \omega_G) = \sum_{q,\alpha} w_q \kappa_{p\alpha|q}, \qquad p(\varnothing \mid \omega_G) = \sum_q w_q (1 - s_q). \tag{78}$$

Their sum is one. In the two-packet nonmaximal state, this simplification is rederived directly rather than inherited from maximal entanglement.

The GNS null left ideal of a vector state on one simple matrix block is not itself a two-sided ideal. The represented kernel, however, is a two-sided ideal and is zero on every active simple block. This is why Section 10 uses the represented kernel and whole inactive blocks for quotient

descent. Replacing N_G by an arbitrary zero-norm subspace would make the quotient statement ill typed.

For dynamics, the one-particle phase in (40) has conjugate e^{-ict} on the first factor and e^{ict} on the second. Hence it cancels in

$$(\overline{u_p(t)} \otimes u_p(t))(\overline{F} \otimes F) = (\overline{F} \otimes F)(\overline{u_q(t)} \otimes u_q(t)). \tag{79}$$

No analogous covariance should be inferred for a source dynamics until its own intertwiner is proved.

D Finite Weyl and cut proof details

For (49), commuting Z^b past $X^{a'}$ gives

$$W(a,b)W(a',b') = \zeta^{ab/2+a'b'/2+ba'} X^{a+a'} Z^{b+b'} = \zeta^{-\sigma(u,v)/2} W(u+v). \tag{80}$$

Orthogonality follows because a nonzero shift has zero diagonal and a nontrivial clock has vanishing character sum. This makes the Weyl operators a basis of the full matrix algebra, so the dressed map is fixed by its action on these generators.

The sector rank in (53) can be counted in the computational basis. For any three slot indices there is exactly one anchor index whose sum has charge q; hence d^3 vectors lie in each sector. On a dressed generator, the three internal shifts acquire phase $\zeta^{tc_{\mathrm{int}}(u)}$ while the anchor shift acquires its inverse. The charge cancels exactly. Since the internal algebra is simple, a nonzero unital representation on any charge block is faithful.

The multiplicativity of the matched-charge cut depends on the declared block algebra. If arbitrary cross-charge operators in $B(\mathcal{H}_C)$ were included, compression would remain CP but would not be a quotient star homomorphism. On $\mathcal{A}_C = \bigoplus_{q_L,q_R} B(\mathcal{H}_L(q_L) \otimes \mathcal{H}_R(q_R))$, the map simply deletes unmatched summands and canonically identifies matched ones. The kernel is therefore exactly (66), and the first isomorphism theorem gives $\mathcal{A}_C/I_{\mathrm{unmatch}} \cong \mathcal{A}_G$.

E Portable verification

The accompanying package contains two standalone NumPy producers, `code/doubled_instrument.py` and `code/weyl_charge_cut.py`, and a small independent diagnostic `code/diagnostic.py`. The first two reproduce the finite values in the main tables, including negative controls. The diagnostic checks the transpose Choi spectrum, observer-restriction witness, failure of generic compression multiplicativity and the Weyl-coordinate dictionary. Running `pythonreproduce.py` from the packet directory needs no source archive, network or model access. Saved outputs and code hashes are part of the packet. The numerical certificates distinguish exhaustive small matrix/label tests from seeded samples. General matrix-level CP follows from the displayed Kraus/isometry proofs; no floating-point tolerance establishes a continuum limit.

Workflow and unresolved attribution

The operator reports that earlier work was led mainly by Fable 5, while results from July 2026 onward were predominantly produced by GPT models. DeepSeek and multiple GPT models also participated in review, checking, preparation and organization. This chronological account is not a complete audited allocation of individual contributions, and a later revision

does not reattribute an earlier result. Human author names and affiliations are intentionally blank. These results have not undergone human expert review. Automated checks and AI reviews are further AI review, not human peer review. Historical source dates, this 29 October 2026 preparation date, and public disclosure dates are distinct. No authorship, ownership, publication-priority or journal-acceptance decision is made. The operator states:

> I genuinely do not know how authorship, intellectual credit, or ownership should be attributed for these results.

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